



Designation: E2033/E2033M – 24

Standard Practice for Radiographic Examination Using Computed Radiography (Photostimulable Luminescence Method)¹

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1. Scope

1.1 This practice covers the minimum requirements for computed radiographic (CR) examination for metallic and nonmetallic materials using X-ray or gamma radiation.

1.2 *Applicability*—The requirements in this practice are intended to control the quality of computed radiographic examinations and are not intended to establish acceptance criteria for parts or materials.

1.3 *Basis of Application*—The requirements of this practice, Practice E2445, and E2446 shall be used together. The requirements of Practice E2445 will provide the baseline performance evaluation and long term stability test procedures for the CR system. Practice E2446 CR performance levels are recommended in Table 1. The user of the CR system shall establish a written procedure that addresses the specific requirements and tests to be used in their application and shall be approved by the Cognizant Radiographic Level 3 before examination of production hardware. The items that shall be determined and addressed in the written procedure are:

- (a) Personnel qualification and certification.
- (b) Minimum effective pixel coverage appropriate to the acceptance criteria and to meet the radiographic image quality level requirements of Table 1.
- (c) Additional tests in accordance with Practice E2445 deemed appropriate.
- (d) Organizations using a gamma source or radiation energy above 320 kV may need to modify the E2445 tests, gauges, or both.
- (e) Maximum allowed unsharpness when other than Table 1.
- (f) The method used to provide image traceability to the part and the examination facility.

1.3.1 This practice also requires the user to perform a system qualification suitable for its intended purpose and to issue a system qualification report (see subsection 7.1).

¹ This test method is under the jurisdiction of ASTM Committee E07 on Nondestructive Testing and is the direct responsibility of Subcommittee E07.01 on Radiology (X and Gamma) Method.

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Additionally, the user shall develop part specific inspection procedures (see subsections 5.5 and 7.5).

1.4 *Units*—The values stated in either SI units or inch-pound units are to be regarded separately as standard. The values stated in each system may not be exact equivalents; therefore, each system shall be used independently of the other. Combining values from the two systems may result in non-conformance with the standard. Where applicable, SI units are shown in brackets [xx].

1.5 *Compliance*—Systems, equipment and materials that do not comply with this practice shall require a waiver from the Cognizant Engineering Organization (CEO).

1.6 *This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.*

1.7 *This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.*

2. Referenced Documents

2.1 ASTM Standards:²

- E746 Practice for Determining Relative Image Quality Response of Industrial Radiographic Imaging Systems below 4 MeV
- E747 Practice for Design, Manufacture and Material Grouping Classification of Wire Image Quality Indicators (IQI) Used for Radiography
- E1025 Practice for Design, Manufacture, and Material Grouping Classification of Hole-Type Image Quality Indicators (IQI) Used for Radiography
- E1030 Practice for Radiographic Examination of Metallic Castings

² For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer Service at www.astm.org/contact. For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.



TABLE 1 Total Image Unsharpness, Maximum

Material Thickness	Maximum Allowed Image Unsharpness (U_{im})	Recommended CR Performance Level in accordance with E2446
≤ 0.5 in. [≤ 12.7 mm]	0.010 in. [0.254 mm]	CR level I or Higher
> 0.5 in. through 1 in. [> 12.7 mm through 25.4 mm]	0.015 in. [0.381 mm]	CR level I or Higher
> 1 through 2 in. [> 25.4 mm through 50.8 mm]	0.020 in. [0.508 mm]	CR level II or Higher
> 2 in. through 4 in. [> 50.8 mm through 101.6 mm]	0.030 in. [0.762 mm]	CR level II or Higher
> 4 in. [> 101.6 mm]	0.040 in. [1.016 mm]	CR level II or Higher

E1032 Practice for Radiographic Examination of Weldments Using Industrial X-Ray Film

E1114 Test Method for Determining the Size of Iridium-192, Cobalt-60, and Selenium-75 Industrial Radiographic Sources

E1161 Practice for Radiographic Examination of Semiconductors and Electronic Components

E1165 Test Method for Measurement of Focal Spots of Industrial X-Ray Tubes by Pinhole Imaging

E1316 Terminology for Nondestructive Examinations

E1647 Practice for Determining Contrast Sensitivity in Radiology

E1735 Practice for Determining Relative Image Quality Response of Industrial Radiographic Imaging Systems from 4 to 25 MeV

E1742 Practice for Radiographic Examination

E1817 Practice for Controlling Quality of Radiological Examination by Using Representative Quality Indicators (RQIs)

E2002 Practice for Determining Image Unsharpness and Basic Spatial Resolution in Radiography and Radioscopy

E2339 Practice for Digital Imaging and Communication in Nondestructive Evaluation (DICONDE)

E2445 Practice for Performance Evaluation and Long-Term Stability of Computed Radiography Systems

E2446 Practice for Manufacturing Characterization of Computed Radiography Systems

E2736 Guide for Digital Detector Array Radiography

E2738 Practice for Digital Imaging and Communication in Nondestructive Evaluation (DICONDE) for Computed Radiography (CR) Test Methods

E2903 Test Method for Measurement of the Effective Focal Spot Size of Mini and Micro Focus X-ray Tubes

2.2 ASNT Standards:³

ANSI/ASNT-CP-189 Standard for Qualification and Certification of Nondestructive Testing Personnel

SNT-TC-1A Recommended Practice for Personnel Qualification and Certification in Nondestructive Testing

2.3 Aerospace Industries Association of America Document:⁴

NAS-410 Certification and Qualification of Nondestructive Testing Personnel

2.4 Society of Motion Picture and Television Engineers (SMPTE):⁵

RP133 Specifications for Medical Diagnostic Imaging Test Pattern for Television Monitors and Hard Copy Recording Cameras

2.5 Government Documents:⁶

NCRP 116 Limitation to Exposure to Ionizing Radiation

NCRP 147 Structural Shielding Design for Medical X-ray Imaging Facilities

2.6 ISO Documents:⁷

ISO 9712 Non-destructive Testing—Qualification and Certification of NDT Personnel

ISO 10012 Measurement Management Systems—Requirements for Measurement Processes and Measuring Equipment

ISO/IEC 17025 General requirements for the competence of testing and calibration laboratories

ISO 17636-2 Non-destructive Testing of Welds—Radiographic Testing—Part 2: X- and Gamma-Ray Techniques with Digital Detectors

ISO 19232-1 Non-destructive Testing—Image Quality of Radiographs—Part 1: Determination of the Image Quality Value Using Wire-type Image Quality Indicators

ISO 19232-2 Non-destructive Testing—Image Quality of Radiographs—Part 2: Determination of the Image Quality Value Using Step/Hole-Type Image Quality Indicators

2.7 EN Documents:⁸

EN 4179 Aerospace Series—Qualification and Approval of Personnel for Non-destructive Testing

EN 12543-2 Non-destructive Testing—Characteristics of Focal Spots in Industrial X-ray Systems for Use in Non-destructive Testing—Part 2: Pinhole Camera Radiographic Method

EN 12543-5 Non-destructive Testing—Characteristics of Focal Spots in Industrial X-ray Systems for Use in Non-destructive Testing—Part 5: Measurement of the Effective Focal Spot Size of Mini and Micro Focus X-ray Tubes.

⁵ Available from Society of Motion Picture and Television Engineers, 3 Barker Ave, White Plains, NY 10601.

⁶ Available from Standardization Documents Order Desk, Bldg. 4 Section D, 700 Robbins Ave., Philadelphia, PA 19111-5094, Attn: NPODS.

⁷ Available from International Organization for Standardization (ISO), ISO Central Secretariat, BIBC II, Chemin de Blandonnet 8, CP 401, 1214 Vernier, Geneva, Switzerland, <http://www.iso.org>.

⁸ Available from British Standards Institution (BSI), 389 Chiswick High Rd., London W4 4AL, U.K., <http://www.bsigroup.com>.



3. Terminology

3.1 Definitions:

3.1.1 Definitions relating to radiographic examination, which appear in Terminology E1316, shall apply to the terms used in this practice.

3.1.2 *1:1, n*—an image display scenario where a single pixel of the image is mapped to a single pixel on the image display monitor.

3.1.3 *Cognizant Radiographic Level 3, n*—the certified level 3 radiographer holding final technical responsibility for the radiographic facility and staff.

3.1.4 *effective pixel size, n*—effective pixel size is equal to the basic spatial resolution of the detector ($SR_b^{detector}$).

3.1.5 *material group, n*—materials that have the same pre-dominant alloying elements and which can be examined using the same material group IQI; a listing of common material groups is given in Practices E747 and E1025.

3.1.6 *pixel coverage, n*—for the purpose of this practice, the term “pixel coverage” refers to the minimum number of effective pixels required to cover a feature such as a critical flaw size or the IQI designated hole or essential wire size, whichever is smallest. Geometric magnification may be required to achieve adequate pixel coverage. Additional information on pixel coverage and geometric magnification can be found in Guide E2736.

4. Summary of Practice

4.1 *Apparatus*—This practice covers application details for computed radiography using a system that consists of a radiation source, a storage phosphor imaging plate (IP) and cassette, an IP scanner/digitizer, scan parameter settings, a workstation with software, an image display monitor, and a digital image archiving system.

4.2 *Safety*—The premises and equipment shall present no hazards to the safety of personnel or property. NCRP 116 and NCRP 147 may be used as guides to ensure that radiographic facilities and procedures are performed so that personnel shall not receive a radiation dose exceeding the maximum permitted by city, state, or national codes.

5. Significance and Use

5.1 This practice establishes the basic parameters for the application and control of the CR examination method. This practice is written so it can be specified on the engineering drawing, specification, or contract.

5.2 *Weld Examination*—Additional information on weld examination may be found in Practice E1032, ISO 17636-2, or both.

5.3 *Casting Examination*—Additional information on casting examination may be found in Practice E1030.

5.4 *Electronic Components*—Radiographic examination of electronic components shall comply with Practice E1161.

5.5 *Part-Specific Examination Technique*—A detailed written procedure in the form of a part-specific examination

technique (7.5) shall be documented for each part, or group of parts, and shall be approved by the Cognizant Radiographic Level 3.

5.6 *Personnel Qualification*—Personnel performing examinations to this practice shall be qualified in accordance with ISO 9712, NAS 410, EN 4179, ANSI/ASNT CP 189, or SNT-TC-1A and certified by the employer or certifying agency as applicable. Other equivalent qualification documents may be used when specified on the contract or purchase order. The applicable revision shall be the latest unless otherwise specified in the contractual agreement between parties.

5.7 *System Qualification*—All CR systems shall be qualified for their intended use. System qualification requirements are specified in subsection 7.1.

5.8 *Process Control*—All CR systems shall be monitored for long term stability (process control) as specified in subsection 7.1.4.

5.9 *Preventative Maintenance*—All CR systems and X-ray machines require periodic maintenance to ensure proper functionality. Preventative maintenance requirements are specified in subsection 6.2.

5.10 *Environmental Conditions*—CR systems should be operated within environmental conditions that are in compliance with manufacturer’s stated acceptable environmental conditions, for example, temperature and humidity. When CR systems are operated outside the specified manufacturer’s environmental ranges, the system shall be qualified (7.1) for such conditions.

6. Equipment and Facilities

6.1 Equipment:

6.1.1 *X-Radiation Sources*—Selection of appropriate X-ray machine parameters (for example, voltage, current, focal spot) is dependent upon the examination requirements for the specimen being examined (for example, material type, geometry, acceptance criteria). The suitability of any X-ray machine and the examination technique parameters shall be demonstrated by attainment of the required radiographic quality level and compliance with all other requirements specified herein.

6.1.2 *Gamma Radiation Sources*—Selection of an appropriate isotope source (for example, energy, source size) is dependent upon the examination requirements for the specimen being examined (for example, material type, geometry, acceptance criteria). The suitability of any gamma ray source and the examination technique parameters shall be demonstrated by attainment of the required radiographic quality level and compliance with all other requirements specified herein.

6.1.3 *Computed Radiography Scanner*—Selection of an appropriate CR scanner is dependent upon the examination requirements for the specimen being examined (for example, material type, geometry, acceptance criteria). The suitability of the CR scanner and the examination technique parameters shall be demonstrated by attainment of the required radiographic quality level and compliance with all other requirements specified herein.



6.1.3.1 The minimum acquisition bit depth of the CR scanner shall be 12.

6.1.3.2 *Discussion*—Guidance for appropriate CR scanner capability may be found in subsection 7.6 and Table 3 for radiographic quality level requirements, Table 1 for recommended CR performance levels, and subsection 7.5 for the examination technique requirements.

6.1.4 *Storage Phosphor Imaging Plate (IP)*—IPs selected for application to this practice shall meet the requirements of the CR scanner manufacturer.

6.1.4.1 *IP Storage and Handling*—IPs should be stored flat (not bent or curled) and in a manner that will not induce pressure marks (for example, vertical storage or storage in non-crushable containers if stacked) when not in use. The IP storage area shall be free of radiation. IPs should be handled with care in order to prevent crimp marks or other surface flaws that will render image artifacts.

6.1.4.2 *IP Cleaning Materials*—Only manufacturer-approved IP cleaning materials shall be used. Care shall be taken to not over-clean IPs in such a way as to damage or remove the clear protective coating.

6.1.4.3 *IP Artifacts*—When IPs are tested for artifacts, compliance with Practice E2445 shall be required. IPs with artifacts that interfere with interpretation in the Area of Interest (AOI) are not acceptable.

6.1.5 *IP Cassettes*—IP cassettes shall be clean, light tight, and constructed of materials that do not interfere with the quality of the radiographic image. Cassettes that exhibit light leaks shall be repaired or discarded.

6.1.6 *IP Cassette Screens*—Front or back metallic screens, or both, should be used with IPs whenever they improve radiographic image quality.

TABLE 3 Radiographic Quality Levels

Radiographic Quality Level	Maximum IQI Thickness, % ^A	Minimum Perceptible Hole Diameter ^{B,D}	Equivalent IQI Sensitivity, % ^{C,E}
1–1T	1	1T	0.7
1–2T	1	2T	1.0
2–1T	2	1T	1.4
2–2T	2	2T	2.0
2–4T	2	4T	2.8

^A $\frac{1}{50}$ th (2 %) of specimen thickness, except the minimum thickness may be 0.005 in. [0.127 mm].

^B The hole, expressed as a multiple thickness of the IQI that shall be clearly visible, except the minimum hole sizes may be 0.010 in. [0.254 mm], 0.020 in. [0.508 mm], and 0.040 in. [1.016 mm].

^C Equivalent IQI (Penetrameter) Sensitivity (EPS) is the thickness of the IQI expressed as a percentage of specimen thickness at which a 2T hole would be clearly visible under the same radiographic conditions—see Practice E1025, Appendix X1.

^D When using Wire Type IQIs, Table 4 of Practice E747 shall be used to determine equivalent wire size to corresponding 1T, 2T, or 4T hole size.

^E EPS values shown above are not applicable for material thicknesses below 0.500 in. [12.7 mm] when using standard Hole Type IQIs with minimum thickness and hole size (Note A & B).

6.1.6.1 Screens shall be visibly free of any cracks, creases, scratches, or foreign materials that would create image artifacts.

6.1.6.2 *Discussion*—For the purpose of this practice, when a metallic screen is used inside the cassette in front of the IP, it shall be referred to as a screen; when used in front of the cassette it shall be referred to as a filter. Back screens, whether inside the cassette or outside the cassette, are considered shielding for backscattered radiation.

6.1.7 *Filters*—Filters should be used at the radiation source whenever they improve radiographic image quality. Filters

TABLE 2 Calibration and Process Control

Check	Method	Frequency	Subsection
Image Display Monitor:			
Brightness	Light Meter	Monthly	6.1.8.1
Contrast	Light Meter	Monthly	6.1.8.2
High Contrast Resolution	Visual	Daily ^A	6.1.8.3
Low Contrast Resolution	Visual	Daily ^A	6.1.8.4
Flicker	Visual	Daily ^A	6.1.8.5
Distortion	Visual	Daily ^A	6.1.8.5
Small Contrast Change	Visual	Daily ^A	6.1.8.6
Light Meter(s)	Calibration	Annual	6.1.10
Image Quality Indicators	Certified	When Procured	6.1.11
	Visual	Prior to Use ^A	6.1.11
Representative Quality Indicators	Visual	Prior to Use ^A	6.1.12
Relative Image Quality Indicators	Certified	When Procured	6.1.13
	Visual	Prior to Use ^A	6.1.13
RIQI Absorber Plate	Certified	When Procured	6.1.13.1
	Visual	Prior to Use ^A	6.1.14
Duplex Wire Gauge	Certified	When Procured	6.1.14
	Visual	Prior to Use ^A	6.1.14
CR Phantom	Certified	When Procured	6.1.15
	Visual	Prior to Use ^A	6.1.15
Reference Standard	Calibration	^B	6.1.16
Measurement Tools	Calibration	^B	6.1.17
Contrast Sensitivity Gauge	Certified	When Procured	6.1.18
	Visual	Prior to Use ^A	6.1.18
Miscellaneous Radiographic Tools	Visual	Prior to Use ^A	6.1.19
Background Ambient Light	Light Meter	6 months ^C	6.3.3.1

^A Documentation of this check is not required.

^B Calibrated and recorded in accordance with ISO/IEC 17025 or ISO 10012, as applicable.

^C Fixed viewing locations with acceptable and controlled ambient lighting conditions need not be re-verified as long as those conditions are maintained.



shall be free of any cracks, creases, scratches, or foreign materials that could create image artifacts.

6.1.8 Image Display Monitor—The image display monitor used for accept/reject evaluations shall be tested in accordance with [Table 2](#) and shall meet the following minimum requirements:

6.1.8.1 Brightness—The minimum brightness at maximum Digital Driving Level (DDL) shall be 250 cd/m²,

6.1.8.2 Contrast—The minimum contrast as determined by the ratio of the brightness at maximum DDL compared to the brightness at the minimum DDL shall be at least 250:1,

6.1.8.3 High Contrast Resolution—The image display monitor shall be capable of displaying linear patterns of alternating pixels at full contrast (modulation depth of 100 %) without aliasing in both the horizontal and vertical directions at the display center and at each of the four corners,

6.1.8.4 Low Contrast Resolution—The image display monitor shall be capable of discriminating linear patterns of alternating pixels at low contrast (1 % modulation patterns where white equals 51 % DDL and black equals 50 % DDL) in both the horizontal and vertical directions at the display center and each of the four corners,

6.1.8.5 Flicker and Distortion—The image display monitor shall be free of screen flicker and discernible geometric distortions, and

6.1.8.6 Small Contrast Changes—The image display monitor shall be capable of displaying a 5 % DDL block against a 0 % DDL background while simultaneously displaying a 95 % DDL block against a 100 % background in a manner that is clearly perceptible to the user.

6.1.9 Image Display Monitor Test Pattern—The test pattern for measuring the image display monitor requirements of subsection [6.1.8](#) shall comply with SMPTE RP 133 and shall be configured to the image display monitors resolution and aspect ratio. The test pattern shall be viewed at 1:1 digital zoom. Alternate test patterns may be used when approved by the Cognizant Radiographic Level 3 provided they include the features described in SMPTE RP 133 required to perform the image display tests specified herein.

6.1.10 Light Meters:

6.1.10.1 Luminance—A calibrated light meter shall be used to measure image display monitors for brightness and contrast and shall measure luminance in candelas per square meter (cd/m²) or foot-lamberts.

6.1.10.2 Illuminance—A calibrated light meter shall be used to measure ambient background lighting and shall measure illuminance in lux (lx), in lumens per square meter (lm/m²) or in foot candles (fc).

6.1.10.3 Calibration frequency for light meters is listed in [Table 2](#).

6.1.11 Image Quality Indicators (IQIs)—A Certificate of Compliance (COC) is required to verify material type and dimensional accuracy. A means to trace COCs to individual IQIs shall be provided. Users shall visually inspect IQIs for damage and cleanliness in accordance with [Table 2](#).

6.1.11.1 Hole-Type IQIs—Hole-type IQIs shall comply with Practice [E1025](#), Practice [E1742](#) Annex A1, or ISO 19232-2, however, the minimum thickness may be 0.005 in. [0.127 mm]

and the minimum hole sizes may be 0.010 in. [0.254 mm], 0.020 in. [0.508 mm] and 0.040 in. [1.016 mm].

6.1.11.2 Wire-Type IQIs—Wire-type IQIs shall comply with Practice [E747](#) or ISO 19232-1 and shall be correlated to hole-type radiographic quality levels.

6.1.11.3 IQI and Shim Material—IQIs and shims shall be of the same material group as the specimen being examined. When IQIs and shims of the same material group are not available, radiographically similar IQIs and shims as defined in Practice [E1025](#) may be used. IQIs and shims of radiographically less dense material than the subject shall be allowed.

6.1.11.4 IQI Shims—Shims used with IQIs shall exceed the IQI dimensions such that the pertinent features of the IQI are visible in the image.

6.1.12 Representative Quality Indicators (RQIs)—When used, RQIs shall comply with the requirements of Practice [E1817](#). Users shall visually inspect RQIs for damage and cleanliness in accordance with [Table 2](#).

6.1.13 Relative Image Quality Indicators (RIQIs)—All RIQIs shall comply with the requirements of Practices [E746](#) or [E1735](#) as applicable, except the material type may be of any material deemed appropriate by the Cognizant Radiographic Level 3. A COC is required to verify material type and dimensional accuracy. Users shall visually inspect RIQIs for damage and cleanliness in accordance with [Table 2](#).

6.1.13.1 RIQI Absorber Plate—The absorber plate shall be of the same material group or radiographically similar material as the RIQIs. Dimensions and surface finish shall comply with Practices [E746](#) or [E1735](#) as applicable. Other thicknesses may be used when approved by the Cognizant Radiographic Level 3. A COC is required to verify material type and dimensional accuracy and surface finish. Users shall visually inspect the RIQI absorber plate for damage and cleanliness in accordance with [Table 2](#).

6.1.14 Duplex Wire Gauge (DWG)—Spatial resolution and unsharpness measurements shall be performed using a DWG that complies with Practice [E2002](#). A COC is required to verify dimensional accuracy. Users shall visually inspect the DWG for damage and cleanliness in accordance with [Table 2](#).

6.1.15 CR Test Phantom—The CR test phantom shall comply with Practice [E2445](#). Either the Type I or the Type II CR test phantom is acceptable for compliance to this practice. A COC is required to verify dimensional accuracy and compliance to [E2445](#). Users shall visually inspect the CR test phantom for damage and cleanliness in accordance with [Table 2](#).

6.1.15.1 In the event a Practice [E2445](#) compliant CR test phantom is not available, other types of gauges or phantoms may be used when approved by the Cognizant Radiographic Level 3 and when applicable, COCs are available to verify dimensional accuracy.

6.1.16 Reference Standard—When image features are measured for accept/reject evaluations, a calibrated physical standard shall be used when calibrating the software measurement tool. Users shall visually inspect the reference standard to ensure the calibration is current and for damage and cleanliness

in accordance with [Table 2](#). A dimensional calibration of the measuring function based upon a verifiable scanned pixel size may also be used.

6.1.17 Measurement Tools—As an alternative to the reference standard, a feature or item included in the image, such as the IQI, may be measured with a calibrated measurement tool (for example, calibrated dial caliper) to establish software calibration. Users shall ensure the calibration is current in accordance with [Table 2](#). A dimensional calibration of the measuring function based upon a verifiable scanned pixel size may also be used.

6.1.18 Contrast Sensitivity Gauge—Contrast sensitivity gauges shall comply with Practice [E1647](#). A COC is required to verify material type and dimensional accuracy. Users shall visually inspect contrast sensitivity gauges for damage and cleanliness in accordance with [Table 2](#).

6.1.19 Miscellaneous Radiographic Tools—Holding fixtures, tooling, and other miscellaneous radiographic aids may be used to aid the radiographic set-up, and shall not interfere with the radiographic image clarity in the AOI or the image of the IQI. Devices used as radiographic set-up aids shall be examined for damage and cleanliness prior to use.

6.2 CR System Preventative Maintenance (PM)—Preventative maintenance shall be performed to ensure the system generates uniform images free of artifacts that would interfere with image interpretation and evaluation in the AOI. Preventative maintenance requirements shall be as stated by the system manufacturer. After PM or repairs, retesting to Practice [E2445](#) is required.

6.3 Facilities—Facilities shall be kept clean and equipped so that acceptable radiographic images are produced in accordance with this practice.

6.3.1 Cleanliness—Environmental conditions involving dust, dirt, or debris shall be controlled to prevent introduction of artifacts to the images and to minimize wear on the CR system hardware.

6.3.2 CR IP Loading and Scanning Area:

6.3.2.1 The surface of the IP loading and scanning area shall be kept visibly clean and free of dust and dirt particles so that contamination is not introduced into the IP cassette or CR scanner

6.3.2.2 Exposed IPs should be handled in conditions of subdued background lighting. It is recommended that exposed IPs be subjected to no more than $2500 \text{ lux} \times \text{seconds}$ of visible light prior to scanning.

6.3.2.2.1 Discussion—The overall exposure to background lighting determines the signal loss and radiographic image quality degradation. Exposure is equal to the lighting intensity multiplied by time.

6.3.3 Image Viewing Area—The image viewing area shall be an area with subdued background lighting with the equipment arranged in such a manner to prevent reflective glare from the surface of the image display monitor. Subdued lighting, rather than total darkness, is required in the image viewing area.

6.3.3.1 Background Ambient Light Level—Background ambient light levels shall not exceed 30 lux (3 fc) at the image display monitor used for image analysis and final disposition. Background ambient light levels shall be measured at the

surface of the image display monitor with the monitor off and recorded in accordance with [Table 2](#).

6.3.4 Environmental Conditions—Humidity and temperature shall be maintained within the limits stated by the equipment and IP manufacturer.

6.4 Software—All software shall be revision controlled. The following software features and tools are required:

6.4.1 Line Profile—A line profile function capable of displaying the pixel values (PVs) along a user defined line as a graph. The line profile tool should also be capable of adjusting the line width where the values of the line profile are averaged across the line width.

6.4.2 Region of Interest Tool—A histogram type tool capable of displaying the PVs of a user-defined Region of Interest (ROI) as a graph. The ROI tool shall also display the size of the ROI (for example, x pixels by y pixels), and as a minimum, the statistical mean and standard deviation of the ROI PVs.

6.4.3 Negative/Positive Image Display—Display images in either negative or positive gray scale (negative or inverse).

6.4.4 Linearized Pixel Values—The software shall be capable of performing calculations using linearized pixel values.

6.4.4.1 Discussion—Linearized pixel values, which are directly proportional to the exposure dose, are required for measuring values such as normalized signal to noise ratio (SNR_N). The linear pixel value is zero if the radiation dose is zero.

6.4.5 Digital Image Magnification (Zoom)—Adjust and display the digital magnification level, as well as display the image at 1:1 pixel mapping (that is, each pixel of the image is mapped to an image display monitor pixel).

6.4.6 Image Pan—Capability to pan the image.

6.4.7 Window Width and Window Level (Window/Level)—Adjust window width (contrast) and window level (brightness).

6.4.8 Size Measurement Tool—Perform measurements for distance or sizing of discontinuities. The software shall be capable of calibrating the measuring tool to a reference standard.

6.4.8.1 Software Calibration—When the system software is capable of self-calibration, such as using pixel size and set up geometry, a reference standard ([6.1.17](#)) may not be required when the software has been validated during system qualification ([7.1](#)).

6.4.9 DICONDE Compliance—Compliance with Practices [E2339](#) and [E2738](#) is recommended.

6.4.10 Image Format—Lossy compression shall not be allowed for images that are used for final product disposition. For systems that are not DICONDE compliant, TIFF images are recommended.

6.4.11 Image Processing Software—Image processing used for final product disposition shall be verified by adequately displaying the pertinent IQI or RQI features. The software shall be capable of saving a copy of the radiographic image with image processing applied.

7. Procedure

7.1 System Qualification—Prior to the examination of production hardware, the CR system shall be tested to establish baseline performance as required in Practice [E2445](#), as well as



its suitability for its intended application. In addition to the Practice E2445 tests, the following minimum tests shall be conducted:

(a) Determine if saturation occurs below the maximum theoretical PV.

(b) Applicable software tools described in Section 6 shall be tested and verified, for example, “the line profile tool shall display the pixel values along a user defined line as a graph;” an example of this requirement can be seen in Practice E2002 Figure 2(b) and Figure 3. A line profile tool that only shows the graph, but does not display the pixel values, would not meet the requirements of subsection 6.4.1.

(c) Image display monitor(s) shall be tested to ensure the requirements of subsection 6.1.8 and 6.1.9 are met.

(d) Background ambient light in accordance with 6.3.3.1.

(e) Image archiving and retrieval shall be tested and verified.

(f) When systems are intended to be used outside of specified environmental conditions, such as portable systems, system qualification tests shall be performed in the expected conditions for temperature and humidity.

7.1.1 System Qualification Test Plan—System qualification shall be documented as a qualification test plan and shall include a complete list of the system components and software (including revision level).

7.1.2 System Qualification Report—The system qualification results shall be documented and the report shall provide traceability to system components and software. The report shall include radiographic techniques for all tests. The report shall include the results of all Practice E2445 baseline performance tests, all qualification test results, results of all image display monitor tests, and background ambient light measurements. For systems that are DICONDE compliant, the report shall also include the system manufacturer’s DICONDE conformance statement. When compliance to a Practice E2446 performance level is required, the report shall include the manufacturer’s Practice E2446 CR System Performance Statement and Characterization Report.

7.1.3 System Retesting—The CR system shall undergo retesting in accordance with Practice E2445 after any repair or preventative maintenance. Additionally, retesting is required after any software updates. Upon moving the CR scanner, whether as a mobile system or for relocation, retesting in accordance with Practice E2445 is recommended.

7.1.4 Long Term Stability (Process Control)—The system shall be periodically retested in accordance with Practice E2445 to monitor long term stability.

7.1.5 Calibration and process control for all other items shall be in accordance with Table 2.

7.1.6 Retest Failure—When any retest failure occurs, the Cognizant Radiographic Level 3 shall determine the appropriate action to take to bring the system into acceptable limits.

7.2 Examination and Coverage—The number of parts examined and the examination coverage of each part shall be as specified by the engineering drawing or other authorizing documentation. When the number of parts to be examined or the amount of coverage is not specified, then all parts shall be examined and shall receive 100 % coverage.

7.3 Examination Sequence—The sequence for radiographic examination shall be as specified by the engineering drawing or other authorizing documentation. When not specified, radiographic examination shall be performed at a stage in the manufacturing process or assembly where pertinent discontinuities can be detected.

7.4 Surface Preparation—Components may be examined without surface preparation or conditioning except as required to remove surface conditions that may interfere with proper interpretation and evaluation of the radiographic images.

7.5 Radiographic Examination Technique—A radiographic examination technique shall be established and documented for each part examined. When the technique is similar for multiple parts, a master examination technique may be used that covers the details common to a variety of parts. The technique shall be capable of consistently producing the detail requirements of this practice and shall be approved by the Cognizant Radiographic Level 3.

7.5.1 The following detail information, as applicable, shall be documented on the radiographic examination technique:

7.5.1.1 Name and address of the examination facility;

7.5.1.2 Customer name;

7.5.1.3 Revision level and date of the technique;

7.5.1.4 Part name and part number;

7.5.1.5 Part material and alloy;

7.5.1.6 Part thickness or thickness range (which the IQI is based on);

7.5.1.7 Required radiographic quality level (Table 3), and details for the compensation principle (7.6.1) if used;

7.5.1.8 Acceptance Requirements—Indicate the criteria by which the components are judged acceptable. Complex components may be divided into zones and separate criteria assigned to each zone in accordance with its design requirements;

7.5.1.9 When used, direct references to ASTM digital reference image standards shall include the grade level for each type of discontinuity permitted for each part or zone;

7.5.1.10 Positioning diagrams, photographs, or sketches shall be made showing the position of the part and the IQI for each view with respect to the radiation source and the IP cassette. The angle of the radiation beam to the part and the IP cassette shall be included. When parts have zones (such as high stress areas), the zones and their acceptance criteria shall be included;

7.5.1.11 Radiographic image identification scheme used to provide traceability of each image to a particular part and view;

7.5.1.12 Radiation Source—For X-ray machines, the X-ray tube manufacturer and model number; for gamma source, the isotope type. All applicable exposure parameters shall be included:

(1) X-ray machine voltage and tube current; or for gamma source, source activity in Curies or gigabecquerel,

(2) Focal spot size or source size corresponding to Test Methods E1114, E1165, or E2903, EN12543-2 or EN12543-5,

(3) Radiation beam filters, collimators, or diaphragms, and

(4) Exposure time.

7.5.1.13 CR scanner make, model, and settings (for example, gain, resolution setting, laser power, laser spot size,

dwell time, scan speed, as applicable) including image processing functions when used during image acquisition;

7.5.1.14 CR IP type and size, including manufacturer;

7.5.1.15 CR IP cassette type (for example, rigid, flexible), and screens when used;

7.5.1.16 Source-to-IP (detector) distance (SDD);

7.5.1.17 Source-to-object distance (SOD);

7.5.1.18 Geometric magnification range of the AOI (min and max);

7.5.1.19 Image unsharpness (U_{im}) of the AOI;

7.5.1.20 The minimum and maximum PV (PV_{min} and PV_{max}) allowed for the AOI;

NOTE 1—When compliance to a Practice E2446 Performance Level is required, the manufacturer's Practice E2446 CR System Performance Statement and Characterization Report will list PV_{min} and PV_{max} . These values are to be observed in terms of overall CR system limits, however, the user may choose to further narrow the PV range allowed as the user's CR system is qualified to a specific application. When compliance to a Practice E2446 Performance level is not required, PV_{max} shall not exceed 90 % of designated saturation value and PV_{min} shall be determined and approved by the Cognizant Level 3 as the minimum pixel value that clearly defines the required features of the IQI or RQI.

7.5.1.21 IQI details (type, material, size, etc.) including shims or blocks when used; or RQI details when used;

7.5.1.22 Basic spatial resolution ($SR_b^{detector}$) of CR scanner and IP combination; and

7.5.1.23 Image review requirements for each view: window width (contrast) and window level (brightness) settings, digital image zoom (minimum and maximum) requirements, and image processing required by the Cognizant Radiographic Level 3 for final accept/reject evaluations. See subsection 7.16.4 for additional details.

7.6 *Radiographic Quality Level*—The radiographic image shall render a clearly defined image of the required IQI features. Table 3 lists the radiographic quality levels. Unless otherwise specified on the engineering drawing or other authorizing documentation, the default radiographic quality level shall be 2-2T.

7.6.1 *Compensation Principle*—If the CR performance level is one level lower than recommended in Table 1, the digital radiograph may be accepted if the required and achieved radiographic quality level of Table 3 is improved by one level. For example, for the radiographic examination of a 3/4 in. plate, the required $U_{im} = 0.015$ in., the recommended CR performance level is I, and the required radiographic quality level is 2-2T. If a CR performance level II system is used, or if the achieved U_{im} is 0.017 in., or both, the digital radiograph may be accepted if a 2-1T radiographic quality level is achieved. This may be achieved by longer exposure time, reduced kV, or both. Use of the compensation principle shall be approved by the Cognizant Level 3 and documented on the radiographic examination technique.

7.7 Image Quality Indicators (IQIs):

7.7.1 *IQI Size Determination*—The IQI shall be based on a thickness no greater than the nominal thickness to be examined. Undersize IQIs may be used when proper thickness IQIs are not available.

7.7.1.1 *Welded Butt Joints*—For welded butt joints on simple structures, such as pipe or plate, except as allowed in

subsections 7.7.2 and 7.7.3, the thickness on which the IQI is based shall be the single wall thickness plus any weld reinforcement allowed. When different size sections are joined, the IQI shall be based on the thickness of the thinner member, unless proper sized IQIs are used on both sides of the weld. Backing strips or rings are not considered part of a weld; therefore, their thickness shall not be included when determining the IQI size.

7.7.1.2 *Other Weld Configurations*—For fillet welds in T-joints or lap joints, weld surfacing, or for joints in complex assemblies where both sides of the joint are not accessible, the thickness on which the IQI is based may be the total thickness through which the radiation beam will pass.

7.7.2 *Double Wall Exposure, Double Wall Viewing*—For double wall exposures with double wall viewing, the thickness of the IQI may be based on a thickness up to the total combined wall thickness being examined.

7.7.3 *Double Wall Exposure, Single Wall Viewing*—For double wall exposures with single wall viewing, the IQI shall be based on the single wall thickness being viewed.

7.7.4 *Other Configurations*—For more complex components, such as castings or assemblies, where multiple wall exposure and multiple wall viewing techniques are used other than as specified in subsections 7.7.2 and 7.7.3, the IQI shall be based on the multiple wall thickness of the component. For multiple wall exposures and single wall viewing techniques, the IQI shall be based on the applicable single wall thickness of the component.

7.7.5 *Propellants/Explosives*—IQIs for radiography of propellants/explosives contained within a vessel shall comply with Practice E1742, Annex A3.

7.8 *IQI Placement*—IQI placement shall be as follows:

7.8.1 An IQI shall be placed on each part examined. When multiple parts are simultaneously examined in a single image, a single IQI may be used at the outer edge of the cone of radiation or the farthest distance from the central beam of radiation.

7.8.2 The IQI shall be placed on an area of the part that will subject it to the same geometric magnification and image unsharpness as the AOI.

7.8.3 When placing the IQI directly on the part being examined, the IQI shall be placed adjacent to the AOI, since accept/reject evaluations cannot be made in the area directly under the IQI.

7.8.4 When it is impractical to place the IQI directly on the part being examined, the IQI shall be placed on the source side of a separate shim or block of the same material or radiographically similar material. When required by contract or the Cognizant Radiographic Level 3, the shim or block and IQI shall be placed on a low absorbing material (such as polystyrene plastic) to ensure that the IQI will be subjected to the same geometric magnification and image unsharpness as the AOI of the part being examined.

7.8.5 IQI placement for welds shall be on the source side, parallel with and at least 0.125 in. [3 mm] but not more than 1.250 in. [32 mm] from the toe of the weld being examined, and placed in such a manner to be at the extreme end of the AOI or the outer edge of the cone of radiation.



7.8.5.1 Dissimilar Materials—When the weld is not radiographically similar to the parent material, or where part geometry precludes placement, the required IQI hole may be placed over the weld, but outside the area of interest. When this is not possible, the IQI may be placed in the AOI and two images shall be taken: one with the IQI, the other without the IQI, both using identical technique and exposure parameters.

(1) When examining welds or components where two or more members of radiographically dissimilar materials (for example, stainless steel to Inconel⁹) are to be viewed on the same image, IQIs of both materials shall be used. When this is not possible, radiographically similar IQIs or IQIs of a radiographically less dense material than the subject may be used.

7.8.5.2 Circumferential Welds—When the radiation source is placed on the axis of the part, and the complete circumference is imaged with a single exposure, at least three equally spaced IQIs shall be used.

7.8.5.3 Weld Junctures—Where portions of longitudinal welds adjoining a circumferential weld are being examined simultaneously, additional IQIs shall be placed on the longitudinal weld at the outer edge of the cone of radiation used to radiograph the circumferential weld.

7.9 Single Exposure IQI—When it is impractical to continually place IQIs on parts requiring more than one exposure using the same radiographic technique (or multiple parts using the same technique), a single exposure taken of the applicable IQIs, both before and after the examination, may be used to qualify the examination series as follows:

7.9.1 When the single IQI method is used, the position of the IQI shall be such to ensure the IQI will be subjected to the same geometric magnification and image unsharpness as the AOI of the part(s) being examined.

7.9.2 Providing the radiographic examination technique is not changed, subsequent exposures may be performed without IQIs.

7.9.3 The single IQI qualification image shall be retaken at least once per shift or when the radiographic examination technique changes.

7.9.4 The examination results shall be invalid when the before or after IQI images fail to demonstrate the required radiographic quality level.

7.9.5 The before and after IQI images shall be considered a part of the digital image data file set for interpretation and archiving.

7.9.6 The single exposure IQI method shall be documented on the radiographic examination technique and approved by the Cognizant Radiographic Level 3.

7.10 IQI Exclusions—IQIs are not required for the following conditions:

7.10.1 Examinations for defect removal, provided the final examination of the area includes an IQI.

7.10.2 Examining assemblies for debris.

7.10.3 Examinations to show material details between two or more materials in assemblies, including honeycomb for detection of fabrication irregularities.

7.10.4 Examining assemblies to show proper assembly, location of internal components, or the presence or absence of materials or components.

7.11 RQIs—RQIs may be used in lieu of IQIs. The use of RQIs shall comply with Practice E1817.

7.12 Technique Calculations—The following measurements and calculations shall be established:

7.12.1 Basic Spatial Resolution of the Detector ($SR_b^{detector}$)—Calculation of U_{Im} and determination of minimum geometric magnification requires measurement of $SR_b^{detector}$ for each CR scanner, scan parameter settings and IP combination. $SR_b^{detector}$ shall be determined in accordance with Practice E2445.

7.12.2 Minimum Geometric Magnification—Geometric magnification may be necessary to ensure the examination technique achieves the required radiographic quality level or detect the required minimum feature size. The cognizant Radiographic Level 3 shall determine the appropriate pixel coverage, using $SR_b^{detector}$ as the effective pixel size to achieve this effect. Where such coverage has not been specified, the geometric magnification should be set to ensure the required IQI hole contains at least a 3×3 effective pixel matrix. The minimum magnification to achieve a minimum effective pixel coverage is determined using Eq 1 for hole-type IQIs or Eq 2 for wire IQIs.

7.12.2.1 Discussion—Typically, Eq 1 or Eq 2 is used to determine the IP and scanner settings that will preclude the use of geometric magnification. For more information on pixel coverage and geometric magnification see subsection 3.1.4.

$$v_{min} = \frac{4.25 \times (SR_b^{detector})}{d} \quad (1)$$

where:

v_{min} = minimum geometric magnification, and,
 d = the diameter of the required IQI hole.

When using wire type IQIs, the minimum geometric magnification shall be set using Eq 2.

$$v_{min} = \frac{2.0 \times (SR_b^{detector})}{wd} \quad (2)$$

where:

v_{min} = minimum geometric magnification, and,
 wd = the essential IQI wire diameter.

7.12.3 Normalized Image Unsharpness (U_{Im})— U_{Im} calculates the effective image unsharpness in the AOI image plane. The maximum allowed image unsharpness is given in Table 1 and may be calculated using Eq 3.

$$U_{Im} = \frac{1}{v} \times \sqrt{U_g^3 + (2.0 \times SR_b^{detector})^3} \quad (3)$$

where:

U_g = $(v-1) \times \phi$
 v = the geometric magnification in the AOI,
 ϕ = the focal spot size or source size.

NOTE 2—Focal spot size or source size corresponding to Test Methods E1114, E1165, E2903, and EN12543-2 or EN12543-5, as applicable.

⁹ Inconel is a trademark of Special Metals Corporation, 4317 Middle Settlement Road New Hartford, NY 13413, <https://www.specialmetals.com>.



7.12.3.1 An alternative method to calculating U_{Im} using Eq 3, is to measure image unsharpness (U_{Im}) in accordance with Practice E2002.

7.13 *Pixel Value*—When compliance to a Practice E2446 performance level is required, PV_{min} and PV_{max} shall be within the values established on the manufacturer's Practice E2446 CR System Performance Statement and Characterization Report for the applicable performance level. When Practice E2446 performance levels are not required, the PV_{min} and PV_{max} shall be as documented on the Level 3 approved radiographic examination technique.

7.14 *Image Acquisition*:

7.14.1 *Minimum Bit Depth*—The minimum bit depth for image acquisition shall be 12.

7.14.2 *Image Identification*—Each image shall be permanently identified to provide traceability back to the component and the radiographic examination report (7.17). The identification may be an image or may be entered as DICOM metadata or other digital metadata. The following minimum information, as applicable, shall be included:

7.14.2.1 Identification and traceability to the component: part name, part number, serial number, examination report number, etc.,

7.14.2.2 Identification of the NDT facility,

7.14.2.3 Date of the examination,

7.14.2.4 View number when location markers are not used, and

7.14.2.5 Repairs shall be identified with R1, R2, R3 and so forth indicating the number of the repair.

7.14.3 *Location Markers*—During radiographic examination, accurate correlation of each part and view/position is essential. To ensure this correlation, location markers shall be used.

7.14.3.1 The images of location markers shall appear on the radiographic image without interfering with image evaluation, and in such an arrangement that it is evident that complete coverage was obtained.

7.14.3.2 *View Numbers*—As an alternative to location markers, view numbers are permitted provided orientation of the image to the AOI can demonstrate the required coverage and location of defects can be found.

7.14.4 *CR IPs*—IPs shall be kept clean so that image artifacts will not be present that interfere with image interpretation and evaluation.

7.14.4.1 IPs should be erased prior to use to prevent ghost images or stray radiation from interfering with subsequent images.

NOTE 3—In some cases, such as when an IP is new or when an IP has not been used in more than 48 h, good practice may require the user to erase the IP more than once prior to use, or to use a secondary erasure cycle. When an IP has been exposed to an exceptionally high radiation dose, the IP may still retain a latent image after standard erasure. In this case, leaving the IP to rest at least one day before re-erasing it may help to remove the latent image.

7.14.4.2 Exposed IPs that are removed from the cassette for scanning, should be handled in conditions of subdued background lighting to prevent degradation of the image.

NOTE 4—It is recommended that exposed IPs be subjected to no more than 2500 lux × seconds of visible light prior to scanning.

7.14.4.3 *IP Cassettes*—IP cassettes shall be visually examined prior to use to ensure they are free of dirt or debris that would create image artifacts that will interfere with image interpretation and evaluation.

7.14.4.4 *Metallic Screens*—Metallic screens should be used with IPs whenever they improve radiographic image quality, IQI sensitivity, or both. Recommended metallic front screen thicknesses and material types are listed in Table 4.

7.14.4.5 *Scattered Radiation Control*—Metallic screens (front, back, or both) or filters should be used as necessary to control scattered radiation from the floor, walls or other surrounding objects from interfering with the image or radiographic image quality requirements. Additionally, scattered radiation effects can be reduced by collimating the radiation beam as much as possible to the AOI.

TABLE 4 Front Metallic Screen Recommendations

NOTE 1—The screen thicknesses listed for the various energy ranges are recommended thicknesses, not required thicknesses. Other thicknesses and materials may be used provided the radiographic quality level is achieved.

NOTE 2—Screens are Pb unless otherwise indicated.

NOTE 3—Pb screens may be replaced completely or partially by Fe or Cu screens. The equivalent thickness for Fe or Cu is three times the thickness of Pb.

NOTE 4— If necessary, the IP should be shielded from backscattered radiation by a sheet of lead of at least 0.010 in. [0.254 mm] thickness or a sheet of tin of at least 0.060 in. [1.5 mm] thickness, placed behind the IP. An additional shielding of steel or copper of 0.005 in. [0.127 mm] should be applied between the lead shield and the IP to reduce the influence of lead X-ray fluorescence radiation. No lead screens should be used in contact with the back side of the IP for radiation energies above 80 keV.

Energy Range / Isotopes	Front Screen, inches	Front Screen, mm
0 MeV to 50 keV	None	None
>50 MeV to 250 keV	0 to 0.005	0 to 0.1
>250 MeV to 1 MeV	0 to 0.010	0 to 0.3
Ir-192, Se-75	0 to 0.020	0 to 0.4
>1 MeV to 5 MeV, Co-60	0.010 to 0.030 Fe or Cu, plus 0.020 to 0.040 Pb	0.3 to 0.8 Fe or Cu, plus 0.6 to 2.0 Pb
>5 MeV	0.020 to 0.150 Fe, Cu or Pb	0.6 to 4.0 Fe, Cu or Pb

**(1) Backscattered Radiation Monitoring and Control**

—The presence of backscattered radiation shall be checked for each new technique by means of a lead letter B positioned behind the IP cassette and in the general area of interest of the digital radiograph. The lead letter B shall be a minimum of ½ in. high and ⅛ in. thick [10 mm in height and 1.5 mm thick]. If there is no image of the lead letter B, then the digital radiograph is acceptable and demonstrates adequate protection against backscattered radiation. If an image of the lead letter B is present in the resulting digital radiograph the following shall apply:

(a) When the image of the B appears as a lower PV than the surrounding background on a digital radiograph, backscattered radiation is present and the digital radiograph shall be considered unacceptable. Re-radiography is required per subsection 7.16.7.

(b) When the image of the B appears as a higher PV than the surrounding background on a digital radiograph, the digital radiograph is acceptable and demonstrates adequate protection against backscattered radiation. Re-radiography in accordance with subsection 7.16.7 would only apply if the image of the B interferes with interpretation in the AOI.

(c) When re-radiography is required due to the presence of backscattered radiation, additional shielding to the IP may be required. The following are recommendations for shielding: 0.010 in. [0.254 mm] thick lead, or, 0.060 in. [1.5 mm] thick tin or copper.

(d) *Discussion*—Lead is not recommended in contact with the back side of the IP at energies above 80 kV due to X-ray fluorescence radiation. A sheet of steel or copper, approximately 0.005 in. [0.127 mm] thick, placed between the IP and lead will reduce this effect.

(2) Testing for backscattered radiation is required for each new radiographic examination technique. The radiographs that were taken during technique development shall be archived for future reference. Additional monitoring is not required during production exposures as long as the original images made during technique development are available for review and as long as the proximity and nature of backscattering objects and exposure conditions are maintained constant as was the case during technique development.

7.15 Image Qualification:

7.15.1 *PV/Dose Range*—Verify the PV range of the AOI is within the PV_{min} and PV_{max} range as established in subsection 7.13.

7.15.2 *Applicable AOI*—Verify IQI coverage as follows:

7.15.2.1 Linear Data:

(1) A single IQI shall represent an area in which the PV range of the AOI is $\pm 15\%$ of the PV measured through the body of the IQI, or, for wire type IQIs, the PV is measured adjacent to the essential wire.

(2) Where one IQI cannot cover the PV range of the AOI within $\pm 15\%$, at least one IQI shall cover the lowest PV of the AOI, and at least one IQI shall cover the highest PV of the AOI to qualify the entire PV range of the AOI.

7.15.2.2 *Logarithmic Data*: A single IQI that is sized for a specific AOI, shall exhibit a PV less than or equal to the PV in the specific AOI.

NOTE 5—For logarithmic data, the 15 % rule in 7.15.2.1 and 7.15.2.2 could allow an IQI to be qualified that is on a much thinner shim or block than the thickness of the AOI, therefore the IQI is only qualified when it is placed on a shim or block of an equal thickness (or thicker) to the AOI.

7.15.3 *Radiographic Quality Level*—The radiographic image shall render a clearly defined image of the required IQI features.

7.15.3.1 For hole-type IQIs, at least two sides and the designated hole shall be visible.

7.15.3.2 For wire type IQIs, the essential wire shall be visible along its full length without interruption within the AOI.

7.15.3.3 When RQIs are used, the required features of the RQI shall be visible as defined and documented according to Practice E1817 requirements.

7.15.4 *Image Identification Verification*—Image identification, including applicable DICOM metadata, shall be verified.

7.16 *Image Interpretation and Evaluation*—All radiographic images shall be examined to identify relevant indications. Relevant indications shall be evaluated to the requirements of the acceptance criteria. Re-radiography shall be required when non-relevant indications or image artifacts are present and interfere with image interpretation.

7.16.1 *Acceptance Criteria*—The acceptance criteria shall be as specified on the engineering drawing or other authorizing documentation

7.16.2 Annotations made to the radiographic image shall be saved in a manner which will not mask or hide the AOI of image.

7.16.3 *Dark Adaptation*—Upon entering the image viewing area to perform image interpretation and evaluation for final product disposition, the image interpreter shall wait at least one minute, or of a sufficiently longer time as needed, before interpreting radiographic images so that the features of the IQI are readily discernible.

7.16.4 Image Viewing Requirements:

7.16.4.1 *Window Width and Window Level (Contrast and Brightness)*—The window width (contrast) and window level (brightness) values as documented on the radiographic examination technique shall be used initially, either with quantitative values or by defined viewing presets. This includes any multiple window width and window level values or viewing presets required due to changes in thickness or material of the component being examined.

7.16.4.2 The interpreter is permitted to adjust both the window width and window level of the image beyond the documented values on the technique provided the required radiographic quality level remains clearly visible.

7.16.4.3 *Digital Magnification (Zoom)*—For the purpose of making measurements, or for situations where the interpreter needs additional zoom for accurate evaluation, the image is permitted to be magnified beyond that specified on the radiographic examination technique for the initial viewing. Digital zoom may be set below 100 % only when the required radiographic quality level remains visible.

7.16.5 *Image Processing*—Since some image processing filters can increase image noise or generate image artifacts, the



interpreter may remove an image processing filter, however, the initial image review shall be performed with the image processing filter applied as required by the radiographic examination technique.

7.16.6 Control of Measurement Accuracy—When indications or features are measured, a system of software calibration is required. Software calibration shall be performed on a known feature, a verifiable pixel size and geometry, or a calibrated reference standard that is imaged at the same geometric magnification plane as that of the feature to be measured.

7.16.7 Re-Radiography—Any image containing image artifacts in the examination AOI shall be re-acquired when the artifact interferes with image interpretation or can be interpreted as a rejectable defect.

7.16.7.1 Whenever there is reasonable doubt as to the interpretation or clarity of the radiographic image due to improper technique or equipment malfunctions, re-radiography shall be required. When the problem cannot be resolved by re-radiography, the radiographic examination technique shall be reviewed by the Radiographic Level 3 and updated as necessary.

(I) When the problem is caused by equipment malfunction, the equipment shall not be returned to service until it is repaired or replaced and re-qualified.

7.16.7.2 Removal of defects identified by radiographic examination shall require re-radiography of the reworked area.

7.16.8 Image Artifact Management—When artifacts are determined to be caused by the IP and result in degradation of image quality or inability to interpret the AOI, the IP shall be cleaned or replaced as necessary. Evidence of slippage, jitter, scan line drop out, fading effects, or combinations thereof, shall be resolved through re-radiography or scanner maintenance.

7.17 Radiographic Examination Reports—The results of all radiographic examinations shall be documented. The report shall reference the acceptance criteria and its revision, provide traceability to the specific parts examined, the disposition of each part (accept/reject), the reason for rejection of any parts, and shall include the name of the interpreter and their level of certification. The following additional information shall be included:

7.17.1 Date of examination,

7.17.2 Name of NDT facility,

7.17.3 Part name, part number, serial number, and view number as applicable,

7.17.4 Examination report number,

7.17.5 CR examination technique number and revision, and

7.17.6 Identification of certified individual(s) performing the examination.

7.18 Marking of Parts—Parts examined by this process shall not be marked in any way that will harm the part, alter material properties, or mask surface features unless specifically required by the CEO. Where marking is required, marking shall be applied in such a manner and location to be harmless to the part. Identification shall not be smeared or obliterated by subsequent handling or processing.

7.19 Retention of Radiographic Data—Radiographic data, including images, examination techniques and examination reports shall be retained in accordance with the provisions of the specific contract. When no specific requirements are specified for records retention, copies of all radiographic data should be delivered with the parts to the purchaser.

7.19.1 Image and Data Archival Requirements—Digital images shall be archived using a reproducible electronic medium.

7.19.1.1 The image archive method shall be documented and verified during system qualification. This shall include traceability to enable retrieval of images at a later date.

7.19.1.2 Archived image files shall maintain the bit depth and spatial resolution of the original image. Lossy compression is not allowed.

7.19.1.3 The original image presented by the CR system shall be preserved without altering the original resolution or pixel values.

7.19.1.4 When image processing is applied (excluding window/level and digital image zoom) to the image used for disposition, the CEO shall approve the method used to ensure that the enhanced image can, if necessary, be retrieved throughout the record retention period.

7.19.1.5 When the single IQI qualification method is used, the single IQI qualification images shall also be archived with the production images.

7.19.1.6 Recording media shall meet the contractual record retention requirements.

7.19.1.7 Digital images and records submitted to customers shall meet all the requirements of this practice.

7.19.1.8 Radiographic examination technique records and examination report records shall also be archived and delivered with the production images.

8. Keywords

8.1 computed radiography; computed radiology; image; image quality indicator; imaging plate; IP; IQI; NDT; nondestructive testing; radiographic testing; scanner; X-ray



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